

Tutorial 3

PS3-11: Bayesian Regression

CS5487 Lecture Notes (2024A)
Prof. Antoni B. Chan
Dept of Computer Science
City University of Hong Kong

$$D = \{(x_i, y_i)\}_{i=1}^N$$

$$y_i = f(x_i, \theta) + \epsilon_i, \quad f(x_i, \theta) = \phi(x_i)^T \theta,$$

↓ look at all data

$$y = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} = \underbrace{\begin{bmatrix} \phi(x_1) & \dots & \phi(x_n) \end{bmatrix}^T}_{\Phi^T} \theta + \epsilon, \quad \epsilon = \begin{bmatrix} \epsilon_1 \\ \vdots \\ \epsilon_n \end{bmatrix}$$

$$y = \Phi^T \theta + \epsilon, \quad \epsilon \sim N(0, \Sigma)$$

Prior $\theta \sim N(0, \Gamma)$

likelihood: $y|x, \theta \sim N(y|\Phi^T \theta, \Sigma)$

(a) Posterior $p(\theta|x, y) = \frac{p(y|x, \theta) p(\theta)}{\int p(y|x, \theta) p(\theta) d\theta} \Leftarrow \text{constant w.r.t. } \theta$

$$\log p(\theta|x, y) = \log p(y|x, \theta) + \log p(\theta) + \text{const.}$$

$$= \log N(y|\Phi^T \theta, \Sigma) + \log N(\theta|0, \Gamma) + \text{const}$$

$$= -\frac{1}{2} \|y - \Phi^T \theta\|_{\Sigma}^2 + \text{const} - \frac{1}{2} \|\theta\|_{\Gamma}^2 + \text{const} + \text{const}$$

$$= -\frac{1}{2} (\cancel{y^T \Sigma^{-1} y} - \cancel{2\theta^T \Phi \Sigma^{-1} y} + \underbrace{\theta^T \Phi \Sigma^{-1} \Phi^T \theta}_{\text{quadratic in } \theta}) - \frac{1}{2} \theta^T \Gamma^{-1} \theta + \text{const.}$$

$$= -\frac{1}{2} \left[\underbrace{\theta^T (\Phi \Sigma^{-1} \Phi^T + \Gamma^{-1}) \theta}_{\text{quadratic in } \theta} - 2 \underbrace{\theta^T \Phi \Sigma^{-1} y}_{\text{linear in } \theta} \right] + \text{const}$$

Trick (PS1-10): Completing the Square:

$$f(x) = x^T A x - 2 x^T b + c$$

$$= (x-d)^T A (x-d) + e = \|x-d\|_{A^{-1}}^2 + e$$

where $\begin{cases} d = A^{-1} b \\ e = c - b^T A^{-1} b \end{cases}$

$$= -\frac{1}{2} \left[\underbrace{\|\theta - (\Phi \Sigma^{-1} \Phi^T + \Gamma^{-1})^{-1} \Phi \Sigma^{-1} y\|_{(\Phi \Sigma^{-1} \Phi^T + \Gamma^{-1})^{-1}}^2}_{\hat{\Sigma}_{\theta}} \right] + \text{const}$$

• log posterior has the form of $-\frac{1}{2} \cdot \text{mahal distance} \rightarrow \text{Gaussian posterior.}$

$$\therefore p(\theta|x, y) = N(\theta|\hat{\mu}_{\theta}, \hat{\Sigma}_{\theta})$$

where $\begin{cases} \hat{\mu}_{\theta} = (\Phi \Sigma^{-1} \Phi^T + \Gamma^{-1})^{-1} \Phi \Sigma^{-1} y \\ \hat{\Sigma}_{\theta} = (\Phi \Sigma^{-1} \Phi^T + \Gamma^{-1})^{-1} \end{cases}$

(*) MAP estimate

$$\hat{\theta}_{\text{MAP}} = \arg \max_{\theta} p(\theta | X, y)$$

$$= (\Phi \Sigma^{-1} \Phi^T + \Gamma^{-1})^{-1} \Phi \Sigma^{-1} y$$

the max is at the mean $\hat{\mu}_{\theta}$

Σ : covariance of obs noise: - inverse weight on the samples



$\Rightarrow$ larger variance $\Rightarrow$ lower weight

$\Rightarrow$ smaller variance $\Rightarrow$ higher weight

Γ : covariance of prior: Γ = "small" $\Rightarrow \Gamma^{-1}$ = large $\Rightarrow \hat{\mu}_{\theta} \rightarrow 0$ (prior mean)

Γ = "large" $\Rightarrow \Gamma^{-1}$ = small $\Rightarrow$ ignore prior (not a strong belief)

(c) $\Gamma = \alpha I, \Sigma = \sigma^2 I$

$$\hat{\theta}_{\text{MAP}} = (\Phi (\sigma^2 I)^{-1} \Phi^T + (\alpha I)^{-1})^{-1} \Phi (\sigma^2 I)^{-1} y$$

$$= (\frac{1}{\sigma^2} \Phi \Phi^T + \frac{1}{\alpha} I)^{-1} \Phi \frac{1}{\sigma^2} y$$

$$= (\Phi \Phi^T + \frac{\sigma^2}{\alpha} I)^{-1} \Phi y = (\Phi \Phi^T + \lambda I)^{-1} \Phi y$$

$$\lambda = \frac{\sigma^2}{\alpha} \geq 0$$

Regularized LS:

$$\hat{\theta} = \arg \min_{\theta} \underbrace{\sum_{i=1}^n (y_i - \phi(x_i)^T \theta)^2}_{\text{data fit term (squared error)}} + \underbrace{\lambda \|\theta\|^2}_{\text{regularizer (keep } \theta \text{ small)}}$$

$$\begin{aligned} &= \arg \min_{\theta} \|y - \Phi^T \theta\|^2 + \lambda \|\theta\|^2 \\ &= y^T y - 2\theta^T \Phi y + \theta^T \Phi \Phi^T \theta + \lambda (\theta^T \theta) \\ &= y^T y - 2\theta^T \Phi y + \theta^T (\Phi \Phi^T + \lambda I) \theta \end{aligned}$$

$$\frac{\partial}{\partial \theta} = -2\Phi y + 2(\Phi \Phi^T + \lambda I) \theta = 0$$

$$(\Phi \Phi^T + \lambda I) \theta = \Phi y$$

$$\hat{\theta}_{\text{LS}} = (\Phi \Phi^T + \lambda I)^{-1} \Phi y$$

left-multiply by $(\Phi \Phi^T + \lambda I)^{-1}$

* Same as MAP soln w/ $\Sigma = \sigma^2 I, \Gamma = \alpha I$